alodi

PROJECT WRITE-UP

Make the second drop land.

An AI agent that rescues failed deliveries for Vietnamese e-commerce. One tap from the courier starts it. The next attempt gets the time, place and person it needs. Decisions about money or a buyer's record stay with people.

Team Develarper · Sea × OpenAI Regional Codex Hackathon, Ho Chi Minh City · 31 October 2026

1 The problem

A failed delivery is a time negotiation that happens off the system.

Shopee buyers cannot choose a delivery window, so buyer and courier settle the time over the phone. When the call is missed, everything that would fix the next attempt shrinks into one error code. The second attempt repeats the first, and the seller pays for the return.

₫40,000	8–12%	0	≈ ₫473M
return fee per failed order, paid by the seller	of orders refused at the door, nationwide	delivery windows a buyer can choose	lost a month in one area of 20 couriers and 4,000 orders a day
Shopee §7.1 · GHN · Sourced	VECOM · Sourced	Shopee 79559 · Sourced	Team cost model · Modelled

2 The solution we built

The courier taps one button. Alodi does the rest, inside Shopee's rules, and hands back structured context.

01	02	03	04	05
Tap	**Reach**	**Extract**	**Decide**	**Redeliver**
No answer, Rescheduled or Refused	Chat at once; calls if silent, only 08:00–20:30	When, where and who, as a context card	Clear answers apply; the rest goes to a person	The courier's next attempt carries the card

WORKING TODAY	WHAT IT DOES
Workflow backend	FastAPI and SQLite: morning availability, courier outcomes, rescue policy, customer-service decisions and a durable scheduler that survives restarts, on 4,000 synthetic orders. 193 automated tests pass, with AI providers simulated.
Voice and memo	GPT-Live-1 over WebRTC holds the buyer call, GPT Transcribe reads the courier's voice memo, and GPT-6.1 Terra extracts the fields. Consent gates, a 180-second call cap and a memo limit are built in. Runs in stub or live mode, through the integrated workspace.
Demo stage	Courier, buyer and operations views side by side, plus an SPX-style hub dashboard. These run a scripted scenario on mock data, so the story stays reliable on stage.

> **Projected value.** At best-in-class rescue rates (vendor benchmark) Alodi would save ₫187.4M a month per area, 40% of the loss, after a running cost of ₫27.3M. Not yet measured. The production design adds Decisions API triage on GPT-6 Luna and local speech and PII services, which are not in this build.

3 Build direction

Primary: **Autonomous & Adaptive AI**. Also: **Deep Domain AI**.

Autonomous & Adaptive AI

An operations agent that runs a rescue from tap to context card without a dispatcher. It adapts to silence, a restart or a low-confidence answer, a policy guard keeps its actions inside hard limits, and it escalates when it should not act alone.

Deep Domain AI

Alodi reflects how last-mile work actually runs: courier outcomes as the trigger, route cut-offs, Shopee's 2–3 contacts within 5 days, permitted calling hours, and the line between asking a buyer and labelling one.

4 How we used Codex

Codex helps specify, build and review. A person merges.

Draft, not for upload. The team confirms this section against what actually happened, then removes this line.

STAGE	HOW CODEX IS USED	SURFACE
Specify	The PRD and business rules become scripts and acceptance checks	ChatGPT + Codex
Build	Pair-programming on the workflow, voice and interface code	Codex CLI · IDE
Parallelise	One task, one branch, one pull request	Codex cloud
Review	Every pull request is reviewed and passes lint, build and test checks in CI before merge	Codex in GitHub
Evaluate	Planned loop: one prompt or schema change per round, kept only if held-out F1 rises. Target F1 ≥ 0.90	Codex + Evals

Team Develarper

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What comes next

Nov 2026	Full build; connect the demo views to the backend
Q1 2027	Shadow mode at one area, no buyer contact
Then	Controlled real channels, then a region
By 2028	National rollout

Sources: Shopee Return & Refund Policy §7.1; Shopee Help Center 79559, 79569; GHN seller guidance; VECOM via Nguoi Lao Dong, 19 Jul 2025; Shipsy NDR benchmark (vendor claim); OpenAI model pages, Oct 2026. Full derivations are in the project report and the numbers PDF.